parent: if references to parents are not stored in children, then each edge is not traversable from child to parent, making it *de-facto* a directed edge.

14.2.3 Acyclic

A *cycle*, in a graph, is a non-empty sequence of edges (u, v_1), (v_1, v_2), …, (v_K, u) that starts and ends at the same vertex.

An *acyclic graph* is a graph that has no cycle.

Both directed and undirected graphs can have cycles, and as such there is a subset of acyclic graphs that is of special interest: *directed acyclic graphs* (*DAG*s).

A DAG has a few interesting properties: there must be at least one vertex that has no incoming edge (otherwise, there would be a cycle); moreover, since it’s acyclic, the set of edges defines a partial ordering on its vertices.